

Math 1127
(Differential and Integral Calculus)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Define continuous function at a point. Discuss the continuity and differentiability at the point $x=0$ of the function, (14)

$$f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right); & x \neq 0 \\ 0 & ; \quad x = 0 \end{cases}$$

- (b) Evaluate $\lim_{x \rightarrow 0} (\cos x)^{\cot^2 x}$. (11)

- (c) If $y = \sin x \sin 2x \sin 3x$, find y_n . (10)

2. (a) State Leibnitz's theorem and use it to prove $(1 - x^2)y_9 = 15xy_8$, (12)
where, $y = \cos(7\cos^{-1}x)$.

- (b) Examine $f(x, y) = x^3 + y^3 - 3axy$ for maximum and minimum values. (11)

- (c) If $U = f(r)$ and $r^2 = x^2 + y^2 + z^2$, then show that $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = \frac{d^2 f}{dr^2} + \frac{2}{r} \frac{df}{dr}$. (12)

3. (a) If $lx + my = 1$ touch the curve $(ax)^n + (by)^n = 1$, then show that (12)

$$\left(\frac{l}{a}\right)^{\frac{n}{n-1}} + \left(\frac{m}{b}\right)^{\frac{n}{n-1}} = 1.$$

- (b) Find all possible asymptotes for the curve (13)

$$x^2 y^2 - 4x^2 y - 5xy^2 - 5x^2 - 6y^2 - 2x + 3y + 2 = 0.$$

- (c) Expand $2x^3 + 7x^2 + x - 1$ in powers of $(x - 2)$. (10)

- 4 (a) State Rolle's theorem. Verify mean value theorem for the function (13)

$$f(x) = (x + 2)(x + 1)(x - 2) \text{ in } -1 \leq x \leq 4.$$

- (b) Differentiate $(\sin^{-1}x)^{\cos x}$ with respect to $(\sin x)^{\ln x}$. (10)

- (c) Define homogeneous function. If $u = \frac{x^2 y^2}{x + y}$, then find the values of (12)

$$(i) x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} \text{ and } (ii) x^2 \frac{\partial^2 U}{\partial x^2} + 2xy \frac{\partial^2 U}{\partial x \partial y} + y^2 \frac{\partial^2 U}{\partial y^2}.$$

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. Integrate the followings:

(a) $\int \sqrt{\frac{x}{x+a}} dx$ (10)

(b) $\int \frac{2\sin x + 3}{5\cos x + \sin x - 1} dx$ (12)

(c) $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$ (13)

6. Evaluate the followings:

(a) $\int_0^{\pi/2} \log \cos x dx$ (12)

(b) $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$. (13)

(c) Evaluate, $\lim_{n \rightarrow \infty} \left[\frac{1}{1+n^4} + \frac{2^3}{2^4+n^4} + \frac{3^3}{3^4+n^4} + \dots + \frac{1}{2n} \right]$. (10)

7. (a) Define Gamma and Beta function. Use Gamma and Beta functions to evaluate $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$. (12)

(b) Evaluate $\int_0^{\infty} \frac{x^8(1-x^6)}{(1+x)^{24}} dx$. (11)

(c) Form a reduction formula for $\int \operatorname{cosec}^n x dx$ and hence evaluate $\int \operatorname{cosec}^5 x dx$. (12)

8. (a) Evaluate $\int_0^{\infty} \frac{\log(1+a^2x^2)}{1+b^2x^2} dx$ by using the rule differentiation under the integral sign. (13)

(b) Evaluate the double integral $\iint (x^2 + y^2) dx dy$ over the positive quadrant to the circle (10)

$$x^2 + y^2 = a^2.$$

(c) Find the area of the segment cut off from the parabola $x^2 = 8y$ by the straight line (12)
 $x - 2y + 8 = 0$.

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2022
Department of Materials Science and Engineering

EEE 1127
(Electrical Engineering Fundamentals)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) "Deflection torque in electrodynamicometer wattmeter is dependent on load power factor for an ac circuit"-Justify the statement with proper diagram. (06)
 - (b) The current flowing through an element is $i = \begin{cases} 2A, & 0 < t < 1 \\ 2t^2, & t > 1 \end{cases}$. Calculate the charge entering from $t=0$ to $t=2s$. (06)
 - (c) State and explain Ampere's Circuital law. Find magnetomotive force around a log straight conductor using Ampere's Circuital law. (11)
 - (d) Define measuring instrument and classify them. How can you extend the range of ammeter and voltmeter. (08)
 - (e) Define magnetomotive force and magnetic field intensity. (04)
2. (a) Deduce the expression for delta to wye conversion. (09)
 - (b) Find the current 'i' in the circuit as shown in figure 2(b). (11)

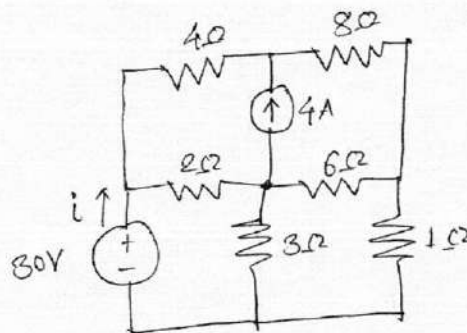


Figure 2(b)

- (c) Using nodal analysis, determine the potential across the 4Ω resistor in figure 2(c). (07)
- (d) Find V_x in figure 2(d) using source transformation. (08)

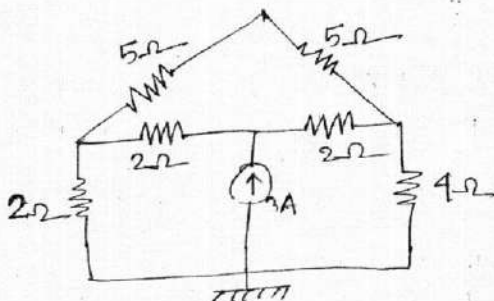


Figure 2(c)

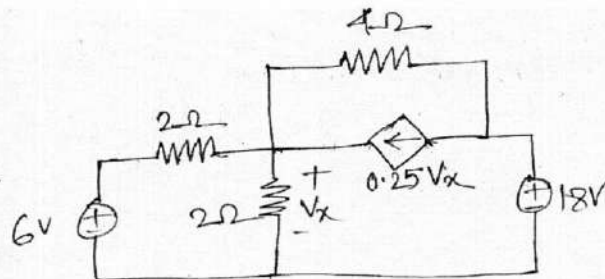


Figure 2(d)

3. (a) State and explain maximum power transfer theorem. (03)

(b) Find the Thevenin equivalent resistance (R_{ab}) and V_{TH} for the following circuit as shown in figure 3(b). (15)

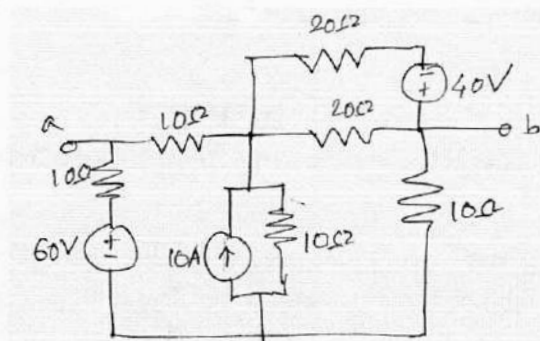


Figure 3(b)

(c) Define RMS value and form factor. Write down the mathematical expression for a 50 Hz sinusoidal voltage of rms value 80V. What is the instantaneous voltage at 45° ? What is the instantaneous value at $t=3\text{ms}$. (07)

(d) Find the impedance and power absorbed by an RL series circuit with proper impedance triangle and vector diagram. (10)

4. (a) What is resonance? Determine the resonant frequency for an ac circuit. (06)

(b) Determine the rms value of the current waveform as shown in figure 4(b). If the current passed through a 2Ω resistor, find the average power absorbed by the resistor. (13)

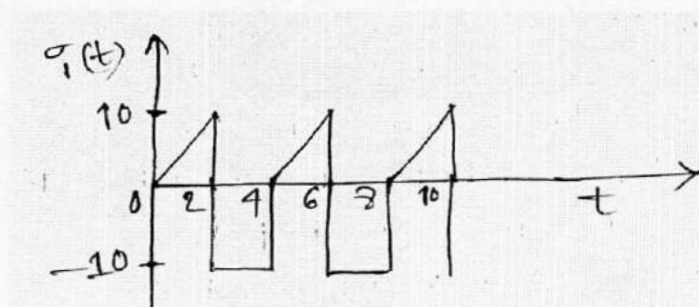


Figure 4(b)

(c) Find the impedance and instantaneous power equation of an RLC series circuit with proper vector diagram. (10)

(d) Find the input impedance of the circuit as shown in figure 4(d). (06)

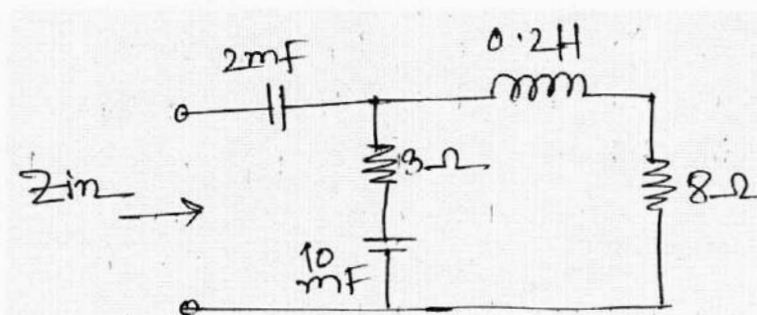


Figure 4(d)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Classify DC generator. Describe the voltage build up process of a DC shunt generator. (10)
- (b) Derive the maximum power development condition of a DC motor and show that the efficiency at this condition drops to 50%. (12)
- (c) Define critical resistance. A short shunt DC generator has armature, series and shunt resistors are 0.05Ω , 0.03Ω and 200Ω , respectively. If the generator supplies 400V at 200A to the load, find the generated emf and armature current of the generator. (Note that 1V drop is allowed for each brushes) (13)
6. (a) What is back emf? Justify that "Back emf self regulates the armature current of DC motor". (12)
- (b) Draw the electrical equivalent circuit of DC shunt, series and compound generators and write down their characteristic equation. (10)
- (c) A belt-driven 100kW, shunt generator, running at 300 rpm on a 220V bus bar continues to run as a motor when the belt breaks, then taking 15 kW. What will be the speed of armature? Given, armature resistance= 0.05Ω and field resistance= 100Ω . Ignore armature reaction. (BCD=1 V/brush). (13)
7. (a) Derive the maximum efficiency condition of a transformer. (10)
- (b) Draw the exact and approximate equivalent circuit of transformer as referred to secondary with their usual meanings. (10)
- (c) Define all day efficiency of a transformer. Classify different types of cooling of transformer. (05)
- (d) Show that induced emf per turn of primary winding and secondary winding is equal for a transformer. (10)
8. (a) Derive the maximum running and starting torque condition of an induction motor. (12)
- (b) Draw the torque-slip characteristics curve of an induction motor. Explain unidirectional torque production in rotor of an induction motor without the aid of commutator. (13)
- (c) Describe the working principle of an Alternator. Mention its advantages and disadvantages and applications. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2022
Department of Materials Science and Engineering

MSE 1101
(Introduction to Materials Science and Engineering)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Explain why materials science and engineering is important with view of historical perspective of evolution of this branch of engineering. (10)
(b) “Materials properties depend on structure and processing technique”-Explain the statement with suitable figure. (13)
(c) Define smart materials. Sketch and describe attractive, repulsive and net force versus interatomic separation for two ions. (12)
2. (a) Calculate the force of attraction between Na^+ and Cl^- ion, the centers of which are separated by a distance of $r_0=2.25$ nm. (10)
(b) Define APF and calculate the APF for Body centered cubic structure. (12)
(c) What is Miller indices? Determine the Miller indices of the following planes shown in figure 2(c). (13)

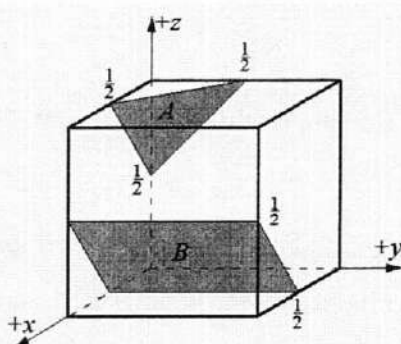


Figure 2(c)

3. (a) Write the names of crystal defects that are found in crystalline solids. Why are defects important in materials? (10)
(b) Calculate the equilibrium number of vacancies per cubic meter for iron at 25°C. The energy for vacancy formation is 0.9 eV/atom, the atomic weight and density (at 25°C) for iron are 55.85g/mol and 7.86 g/cm³, respectively. (12)
(c) Explain the role of defects in failure. Differentiate between transgranular and intergranular fracture with suitable diagram. (13)
4. (a) Derive the relationship between engineering stress and true stress. Hence, explain why engineering stress is used in conventional process instead of true stress in a material. (13)
(b) Describe the stages of ductile fracture and point out the differences with brittle fracture. (12)
(c) Write short note on (i) Creep, (ii) Fatigue. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) "Plastic deformation can be applied to make a material stronger"-Explain this statement. (10)
- (b) Aging can strengthen a material but overaging can reduce it. Explain this phenomenon of hardening by choosing a suitable example with diagram. (15)
- (c) Explain transformation toughening mechanism of partially stabilized Zirconia by Al_2O_3 or ZrO_2 . (10)
6. (a) What are composite materials? Explain why length and diameter of fibers play vital roles for the enhancement of load bearing capabilities of composite materials with detailed sketches. (10)
- (b) Explain with necessary example, how fracture toughness of ceramic matrix composite can be improved. (07)
- (c) A continuous and aligned fibrous composite having a cross-sectional area of 900 mm^2 is subjected to an external tensile load. If the stress sustained by the matrix and fiber phase is 200 MPa and 5 MPa respectively, the force sustained by the fiber phase is 70000 N and the total longitudinal composite strain is 1.5×10^{-5} , then determine:
- (i) The force sustained by the matrix phase.
- (ii) The modulus of elasticity of the composite material in the longitudinal direction.
- (iii) The moduli of elasticity for fiber and matrix phases. Given, $\sigma_c = \sigma_m V_m + \sigma_f V_f$,
and $E_c = E_m V_m + E_f V_f$.
7. (a) What is Fermi energy? Explain with band diagram, the difference between metal, semi-conductor and insulator. (12)
- (b) A 10 cm diameter cylindrical silicon specimen 60 cm long in which a current of 0.5 A passes in an axial direction. A voltage of 24 V is measured across two probes that are separated by 450 mm . Compute the electrical conductivity and resistance over the whole specimen. (13)
- (c) How does temperature affect the conductivity of intrinsic and extrinsic semiconductors? (10)
8. (a) Discuss the optical phenomena within solid when light interacts with solid. (10)
- (b) Explain the working principle of LASER. (10)
- (c) Why is NDT important in chemical industries? A conductor has to be tested for any flaws either surface or subsurface. Choose a suitable NDT technique to detect the flaw and explain the technique with diagram. (15)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2022
Department of Materials Science and Engineering

Ph 1127
(Optics and Waves)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Two simple harmonic vibrations acting at right angles to each other have the time period in the ratio 1:2. The phase difference between the two vibrations is $\pi/2$. Show graphically that the resultant curve is parabola. (12)
(b) What is logarithmic decrement of a damped oscillatory motion? How damping co-efficient can be determined of such a system? (13)
(c) The force and displacement of a simple dynamic system under going sinusoidal excitation are given by the equations $F = 10\sin\left(\frac{\pi t}{10}\right) N$ and $y = 0.10\sin\left(\frac{\pi t}{10} - \frac{\pi}{3}\right) m$. (10)
Calculate the work done by the excitation force in (i) 10 seconds and (ii) 2.5 minutes.
2. (a) Show that the energy of a plane progressive wave is given by $E = 2\pi^2 \rho n^2 a^2$ (10)
Where, the terms have their usual meanings.
(b) Discuss mathematically the formation of stationary waves and explain how the characteristics change with (i) time and (ii) distance. (15)
(c) Plane harmonic waves of frequency 500Hz are produced in air with displacement amplitude $10^{-3}cm$. Deduce (i) the pressure amplitude, (ii) energy density and (iii) energy flux in the wave. (10)
3. (a) What is meant by ultrasonic waves? Mention the properties of ultrasonic waves. (10)
(b) What is Doppler effect? Find the expression for the apparent frequency of the note when the source and listener are (i) moving towards each other and (ii) moving away from each other. (15)
(c) Two aeroplanes A and B are approaching each other and their velocities are 650km/hr and 850km/hr respectively. The frequency of the note emitted by A as heard by the passengers in B is 1100Hz. Calculate the frequency of the note as heard by the passengers in A. Velocity of sound in air is 332m/s. (10)
4. (a) Define the intensity and energy density at a point in a plane wave of sound. Obtain an expression for the intensity of a plane wave in terms of acoustic pressure level. (13)
(b) What are the acoustic requirements of a good auditorium? Discuss the factors influencing loudness. (12)
(c) An air conditioning unit operates at a sound intensity level of 75db. If it is operates in a room with an existing sound intensity level of 70db, what will be the resultant intensity level? (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What is compound microscope? Derive an expression for the magnification factor of a compound microscope. (15)
- (b) What is achromatism? Derive the condition for achromatism of two lenses placed in contact. (10)
- (c) Two thin converging lenses of powers 5D and 4D are placed co-axially 10cm apart. Find the focal length of the combination. (10)
6. (a) What is Huygens wave theory of light? What is meant by a wavefront? How is it produced? (10)
- (b) Describe Newton's ring experiment and explain how it can be used to determine the wavelength of sodium light. (15)
- (c) A Fresnel bi-prism having an angle of 1° and refractive index 1.61 forms interference fringes on a screen placed 80cm from the prism. If the distance between the source and bi-prism is 20cm, find the fringe separation when the wavelength of light used is (i) $6800\text{\AA}$ and (ii) $5890\text{\AA}$. (10)
7. (a) What is Fraunhofer diffraction? Derive an expression for the intensity distribution due to Fraunhofer double slit diffraction. (15)
- (b) Discuss the missing order in a double slit diffraction pattern. (10)
- (c) A plane transmission grating having 3000 lines per cm gives an angle of diffraction of a ray of light by 30° in the third order. Calculate the wavelength of light. (10)
8. (a) Show that the light waves are transverse waves and not longitudinal waves. (10)
- (b) Discuss the constructional details and working principles of a Nicol Prism. How can it be used as a polarizer and an analyzer? (15)
- (c) Calculate the specific rotation of the given sample of sugar solution if the plane of polarization is turned through 26.4° . The length of the tube containing 20% sugar solution is 20 cm. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2022
Department of Materials Science and Engineering

Ch 1127
(Inorganic and Physical Chemistry)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Draw the Lewis structures for the following species: i) NH_3 ii) SO_4^{2-} iii) O_3 . (06)
(b) Draw and explain the molecular orbital energy level diagram of O_2^+ . (12)
(c) Explain the inter and intra-molecular hydrogen bonding with examples. (09)
(d) What is lattice energy and what role does it play in the stability of ionic compounds? (08)
2. (a) Explain an electrochemical cell of Galvanic type and electronic type. Write down the sign convention and types of reaction of these cells. (09)
(b) Derive an expression for Nernst equation. (10)
(c) Determine the feasibility of the reaction (06)
$$\text{Al(s)} + 3\text{Sn}^{4+}(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{Sn}^{2+}(\text{aq})$$

Given, $E_{\text{Al}/\text{Al}^{3+}}^\circ = -1.66\text{V}$ and $E_{\text{Sn}^{4+}/\text{Sn}^{2+}}^\circ = 0.15\text{V}$.

(d) How pH of a solution is determined by using Quinhydrone electrode? (10)
3. (a) Distinguish between electronic and electrolyte conduction. (07)
(b) What is transport number? Describe one method of determining transport number. (11)
(c) What is Salt bridge and why it is used? (08)
(d) The equilibrium constant for the reaction $\text{Sr(s)} + \text{Mn}^{2+}(\text{aq}) \rightleftharpoons \text{Sr}^{2+}(\text{aq}) + \text{Mn(s)}$ is 2.69×10^{12} (09)
at 25°C . Calculate E° for a cell made up for Sr/Sr^{2+} and Mg/Mg^{2+} half cells.
4. (a) Define instantaneous rate, rate law and order of reaction. (09)
(b) What is activated complex? Discuss the collision theory of reaction rate. (10)
(c) For the reaction $2x + y \rightarrow L$, find the rate law from the following data. (08)

[x]/min	[y]/min	[rate]/ MS^{-1}
0.2	0.02	0.15
0.4	0.02	0.30
0.4	0.08	1.20

(d) Why high molecularity reaction are rare? Explain with diagram. (08)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) State Raoult's law. Derive the mathematical expression of Raoult's law. (11)
- (b) What is ideal solution? Draw a diagram for the change of vapour pressure of a (12)
- i) positively deviated non-ideal solution with composition
- ii) negatively deviated non-ideal solution with composition.
- (c) What are colligative properties? Draw the phase diagram showing the effect of a nonvolatile solute on freezing point and boiling point. (12)
6. (a) Define an acid and a base according to the Brønsted-Lowry concept. Give an acid-base equation and identify each species as an acid or base. (12)
- (b) In the following reactions, identify the Lewis acid and the Lewis base. (12)
- (i) $\text{Ag}^+ + 2\text{NH}_3 \rightleftharpoons \text{Ag}(\text{NH}_3)_2^+$
- (ii) $\text{B}(\text{OH})_3 + \text{H}_2\text{O} \rightleftharpoons \text{B}(\text{OH})_4^- + \text{H}^+$
- (iii) $\text{O}_2^- + \text{CO}_2 \rightleftharpoons \text{CO}_3^{2-}$
- (c) State the phase rule. Explain the terms involved in it. (11)
7. (a) In the following equilibrium $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, how many number of phases, components and degrees of freedom are available? (09)
- (b) State and derive Phase rule thermodynamically. (10)
- (c) Discuss the simple eutectic system of Silver-Lead (Ag-Pb) with a neat diagram. (10)
- (d) Explain the following terms giving examples- i) Invariant system ii) Congruent melting point. (06)
8. (a) How does a sol differ from a gel? (06)
- (b) How can you prove that colloidal particles are electrically charged? How do the colloidal particles acquire electrical charge? (12)
- (c) Why are lyophilic colloidal sols more stable than lyophobic colloidal sol? (08)
- (d) Write a short note on micelle. (09)